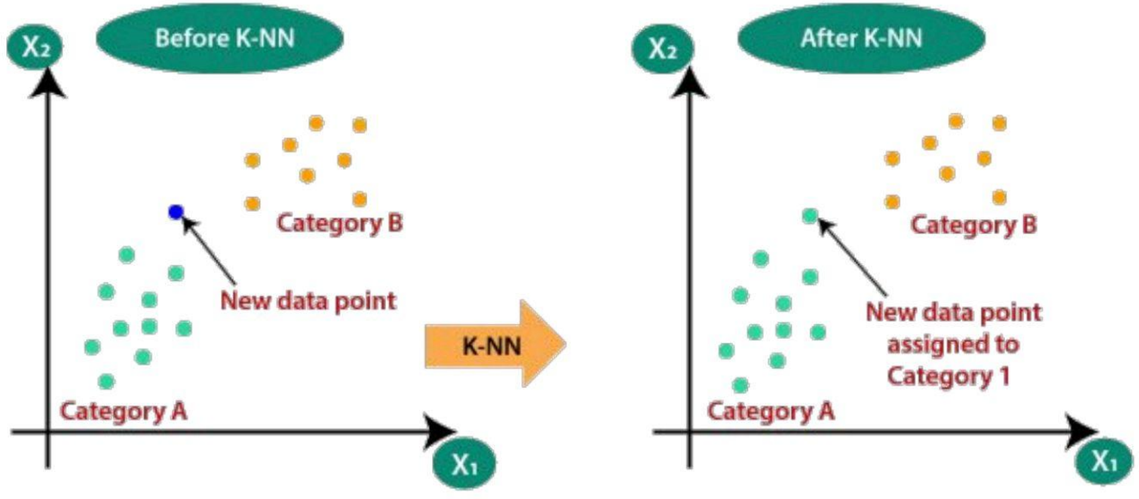
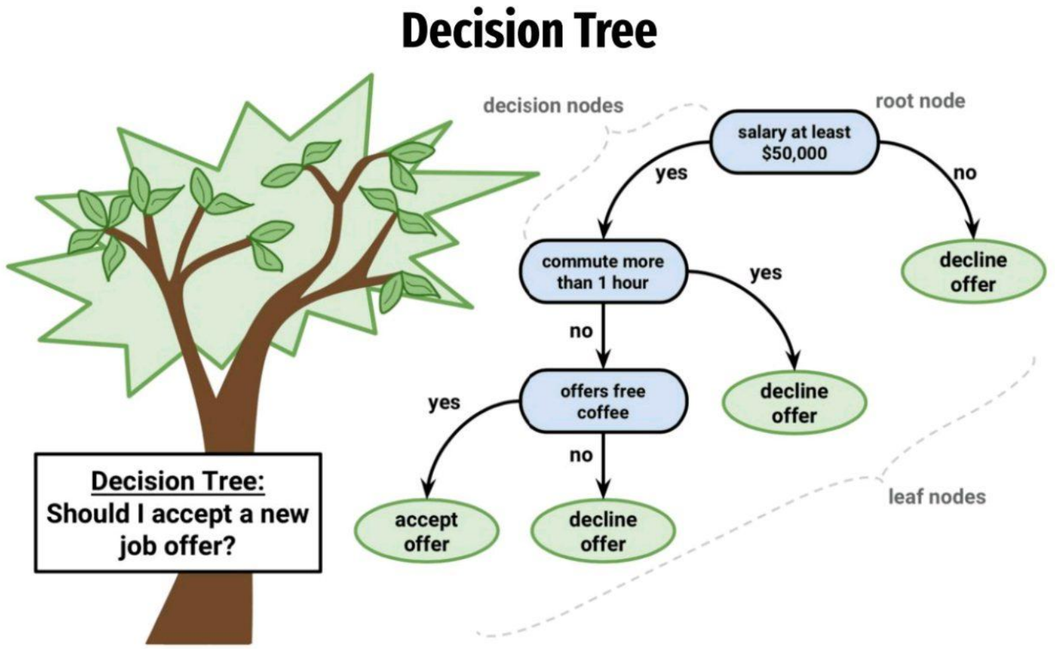
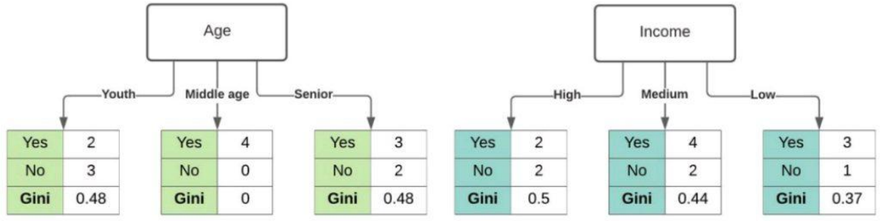
K-Nearest-Neighbour



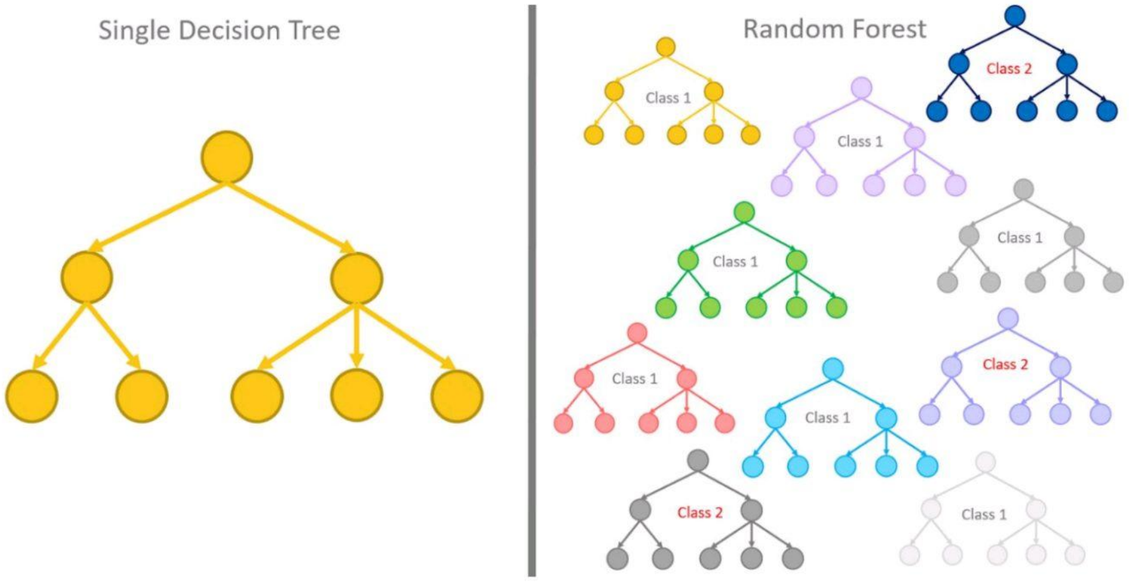

Decision Tree
decision nodes
root node
salary at least $50,000
yes
no
commute more than 1 hour
yes
decline offer
no
offers free coffee
yes
decline offer
no
accept offer
decline offer
leaf nodes
Decision Tree:
Should I accept a new job offer?

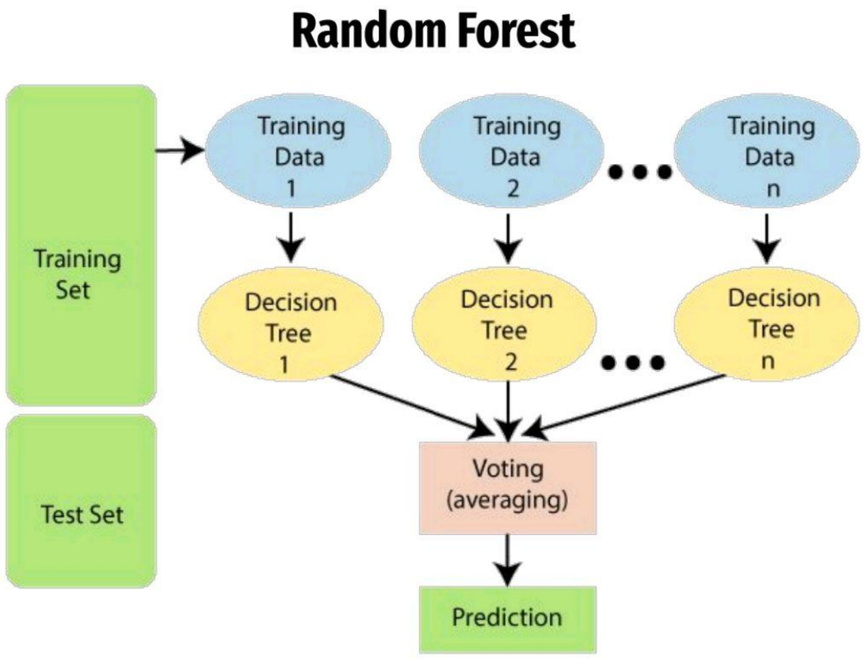
Decision Tree

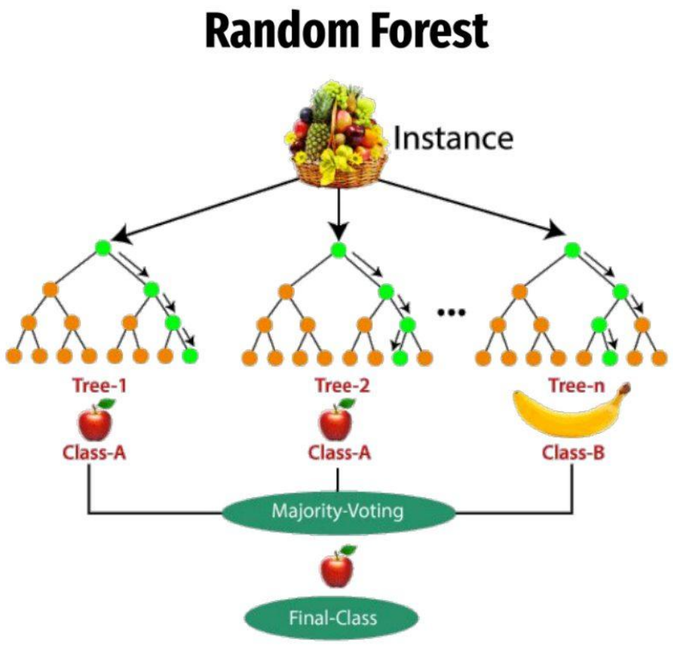
$$Gini = 1 - \sum_{i=1}^{C} (p_i)^2$$



Random Forest




Random Forest
Training Set
Test Set
Training Data 1
Training Data 2
Training Data n
Decision Tree 1
Decision Tree 2
Decision Tree n
Voting (averaging)
Prediction


Random Forest
Instance
Tree-1
Tree-2
Tree-n
Class-A
Class-A
Class-B
Majority-Voting
Final-Class